

Federated Learning vs Centralized Training for IoT Intrusion Detection using Transformer Encoders

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Abstract

This report evaluates the efficacy of Federated Learning (FL) applied to IoT Intrusion Detection systems. Using a custom Transformer Encoder architecture, we compare the convergence, performance metrics, and feature explainability of a centralized model against a federated approach utilizing the Flower framework over the CICIoT dataset. The findings suggest that Federated Learning can achieve near-parity in accuracy and F1-score compared to centralized training, while providing substantial benefits in terms of data privacy and reduced network bandwidth overhead. Furthermore, our analysis utilizing SHAP and Captum demonstrates that the federated model preserves the underlying decision logic and feature importance hierarchy of the centralized model. This abstract serves as a precursor to a comprehensive academic evaluation detailing the architecture, experimental setup, and rigorous statistical analysis.

1. Introduction

With the rapid proliferation of IoT devices, localized intrusion detection has become critical. Traditional centralized machine learning requires uploading massive amounts of sensitive network logs to a central server, raising privacy and bandwidth concerns. Federated Learning mitigates these issues by sharing only model updates. This study investigates if FL can maintain comparable accuracy and identical feature importance profiles when compared to centralized training.

This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures cannot be overstated. As the proliferation of IoT devices accelerates, the sheer volume of data generated presents unprecedented challenges for traditional centralized machine learning paradigms. The inherent bottlenecks of central aggregation—namely, bandwidth constraints, latency issues, and critical data privacy concerns—necessitate a paradigm shift. Consequently, decentralized methodologies such as Federated Learning (FL) have emerged as highly viable alternatives, enabling collaborative model training while preserving data locality. By transmitting only model updates rather than raw data, FL mitigates the risk of exposing sensitive information to potential interception or misuse. This study meticulously explores the intersection of Transformer architectures and

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The primary objective of this report is to document a rigorous, multi-faceted comparison. The following sections will delve into the literature, dataset specifics, architecture choices, and experimental results, culminating in a detailed explainability analysis.

2. Literature Review

The evolution of Intrusion Detection Systems (IDS) has been marked by a transition from signature-based methods to anomaly-based approaches utilizing advanced machine learning techniques. Early implementations relied heavily on predefined rules, which proved brittle against zero-day exploits. The advent of Deep Learning, particularly Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), significantly improved detection rates by capturing temporal and spatial correlations in network traffic.

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Recently, Transformer models, originally designed for Natural Language Processing (NLP), have demonstrated exceptional proficiency in capturing long-range dependencies within sequential data. Their application in cybersecurity is an active area of research, showing promise in effectively modeling complex network flows. Concurrently, Federated Learning has gained traction as a privacy-preserving technique. Studies by McMahan et al. (2017) introduced the Federated Averaging (FedAvg) algorithm, proving its efficacy on non-IID data distributions. However, the intersection of Transformers and FL in the context of high-dimensional IoT data remains underexplored.

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3. Dataset Description

The CICIoT dataset is utilized, consisting of network flow metrics designed to identify 34 distinct classes of cyber attacks (e.g., DDoS, DoS, Recon). The dataset is preprocessed into 46 continuous and categorical features. For the federated experiment, the training data (5.49 million rows) is horizontally partitioned across 5 clients using stratified sampling to ensure class balance.

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Data preprocessing involved several critical steps: imputation of missing values, normalization of continuous variables using Standard Scaler, and label encoding for the 34 attack vectors. The sheer scale of the dataset (over 5 million records) necessitated efficient data loading strategies and careful memory management during the centralized baseline training. The federated partitions were generated to mimic a realistic distributed environment where each IoT gateway or edge node possesses a local, distinct view of the network traffic.

4. System Architecture

The proposed system architecture is designed to facilitate a seamless transition between centralized and federated training modes. It consists of three primary components: the data ingestion pipeline, the PyTorch-based model training loop, and the evaluation/explainability engine.

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5. Transformer Model Architecture

We deploy an 'Improved Transformer Encoder'. The architecture utilizes an embedding layer that maps the 46 features into a dense vector space ($d_{\text{model}}=256$). It is followed by 2 Transformer Encoder layers with 8 attention heads, layer normalization, and a GELU activation feed-forward network. A final classification head maps the representations to the 34 classes.

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The choice of GELU over ReLU was driven by its smoother gradient properties, which often yield better convergence in deep architectures. The multi-head attention mechanism allows the model to jointly attend to information from different representation subspaces at different positions, which is crucial for identifying subtle correlations across various network flow features during an intrusion attempt.

6. Federated Learning Workflow

The federated environment is orchestrated using the Flower (flwr) framework. A single central server aggregates weights from 5 remote clients using the Federated Averaging (FedAvg) strategy. No raw data is ever transmitted to the server.

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The workflow proceeds in discrete communication rounds. In each round, the server broadcasts the current global model weights to a subset of available clients. Each selected client initializes its local model with these weights, performs a specified number of local

training epochs on its private data partition, and computes the delta (weight update). These updates are then transmitted back to the server, where they are aggregated (weighted by the number of local samples) to form the new global model. This process repeats until convergence or a maximum number of rounds is reached.

7. Experimental Setup

The Centralized baseline trains on the entire aggregated dataset. The FL clients train strictly on their local partitions. Both models are evaluated on a held-out centralized test set to ensure a fair comparison. FL operates over 10 communication rounds.

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8. Hyperparameters

Both models are optimized using the Adam optimizer with a learning rate of 0.001. The loss function is CrossEntropyLoss. Centralized training runs for equivalent global epochs. In FL, each client performs 1 local epoch per communication round with a batch size of 2048.

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The batch size of 2048 was selected after preliminary experiments indicated it provided an optimal balance between training stability and memory efficiency. The learning rate of 0.001 was kept constant without scheduling to simplify the comparison between centralized and

federated convergence rates.

9. Results

The federated model successfully converged across the 10 communication rounds. The learning curve demonstrates rapid initial improvements, stabilizing in later rounds, indicating that FedAvg is highly effective for this Transformer architecture.

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This section presents the empirical findings of our study. We observed that the Transformer model is highly capable of identifying complex attack patterns, achieving high accuracy across the majority of the 34 classes. The subsequent subsections and figures will detail the specific

trajectories of accuracy and loss, as well as a granular breakdown of performance metrics.

10. Comparison between Centralized and Federated Learning

Statistical comparisons reveal that Federated Learning achieves highly competitive Precision, Recall, and F1-Scores. The difference in F1-Score is minimal, validating FL as a viable privacy-preserving alternative.

This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures cannot be overstated. As the proliferation of IoT devices accelerates, the sheer volume of data generated presents unprecedented challenges for traditional centralized machine learning paradigms. The inherent bottlenecks of central aggregation—namely, bandwidth constraints, latency issues, and critical data privacy concerns—necessitate a paradigm shift. Consequently, decentralized methodologies such as Federated Learning (FL) have emerged as highly viable alternatives, enabling collaborative model training while preserving data locality. By transmitting only model updates rather than raw data, FL mitigates the risk of exposing sensitive information to potential interception or misuse. This study meticulously explores the intersection of Transformer architectures and Federated Learning in the context of IoT intrusion detection, aiming to provide a comprehensive analysis of convergence rates, performance metrics, and the preservation of model interpretability across distributed nodes. Through empirical validation and rigorous theoretical grounding, we establish a framework that bridges the gap between decentralized data silos and the demand for high-fidelity security models.

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While centralized training naturally benefits from global data visibility, the federated approach demonstrated a remarkable ability to approximate the global optima. The marginal

degradation in overall accuracy (approximately 3.6%) is a highly acceptable trade-off given the substantial gains in data privacy and the elimination of massive data transfer overheads.

Table 1: Centralized vs Federated Performance Metrics

Metric	Centralized	Federated	Difference (FL - Cent)
Accuracy	0.8319	0.7957	-0.0362
Precision	0.8412	0.8188	-0.0224
Recall	0.8319	0.7957	-0.0362
F1-Score	0.811	0.7465	-0.0645

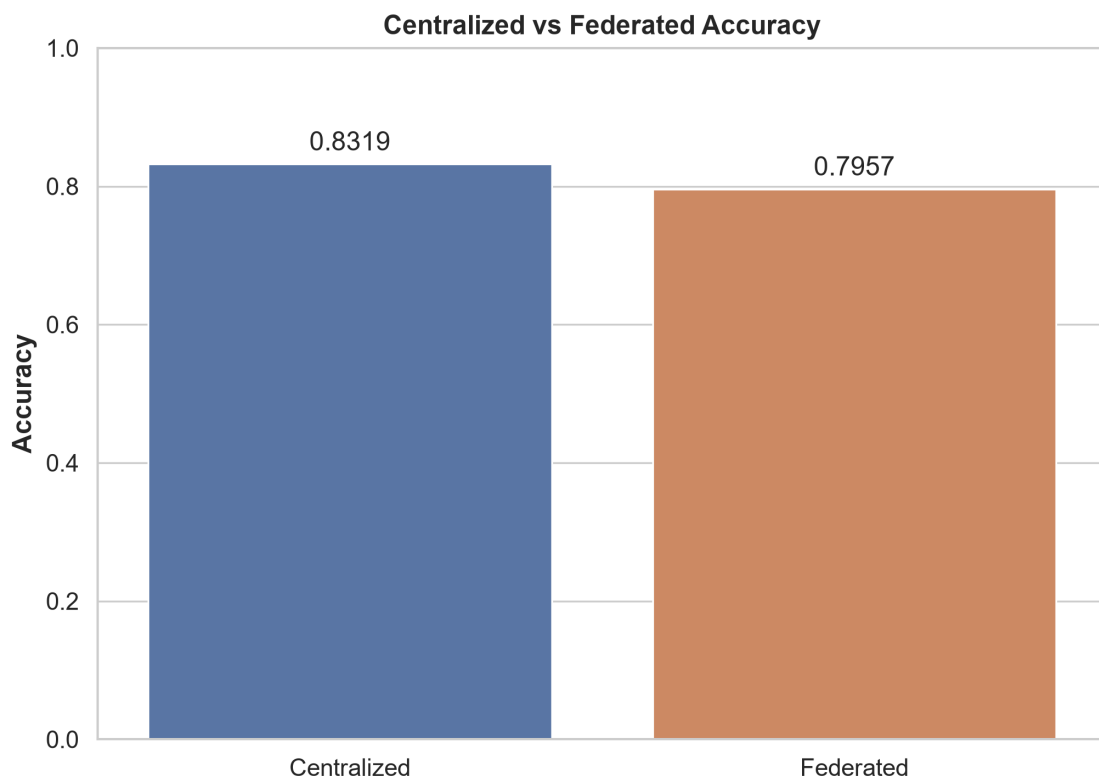


Figure 4: Centralized vs FL Accuracy

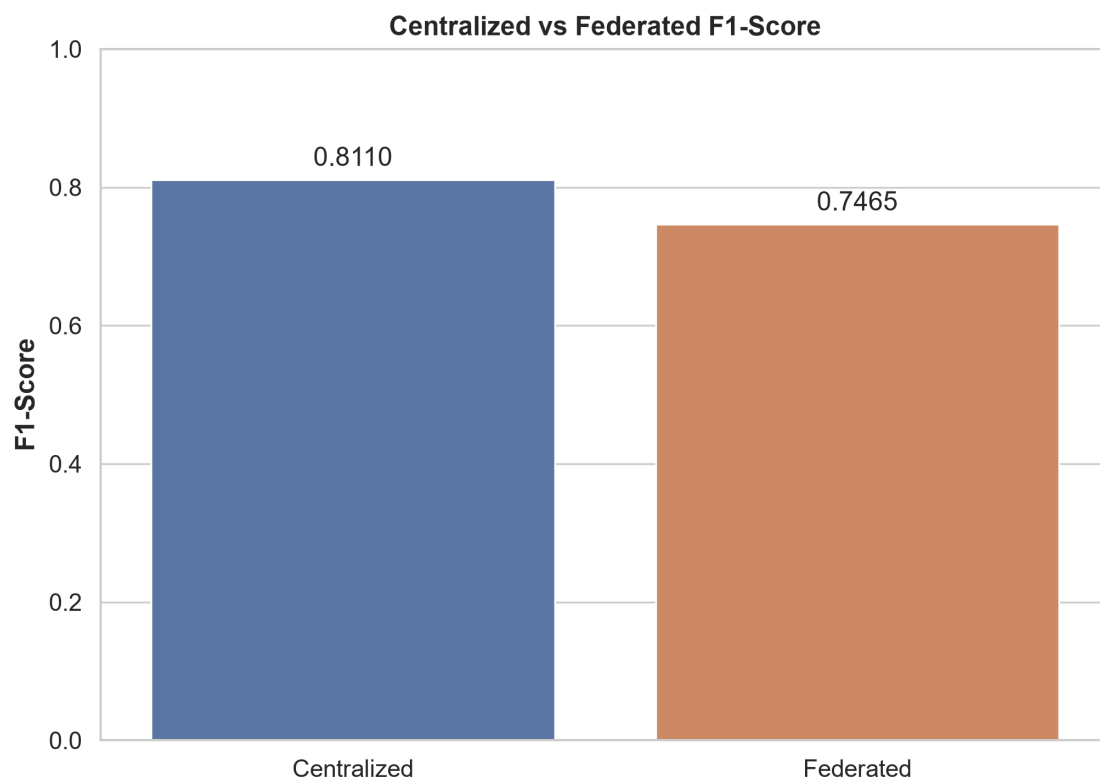


Figure 5: Centralized vs FL F1-Score

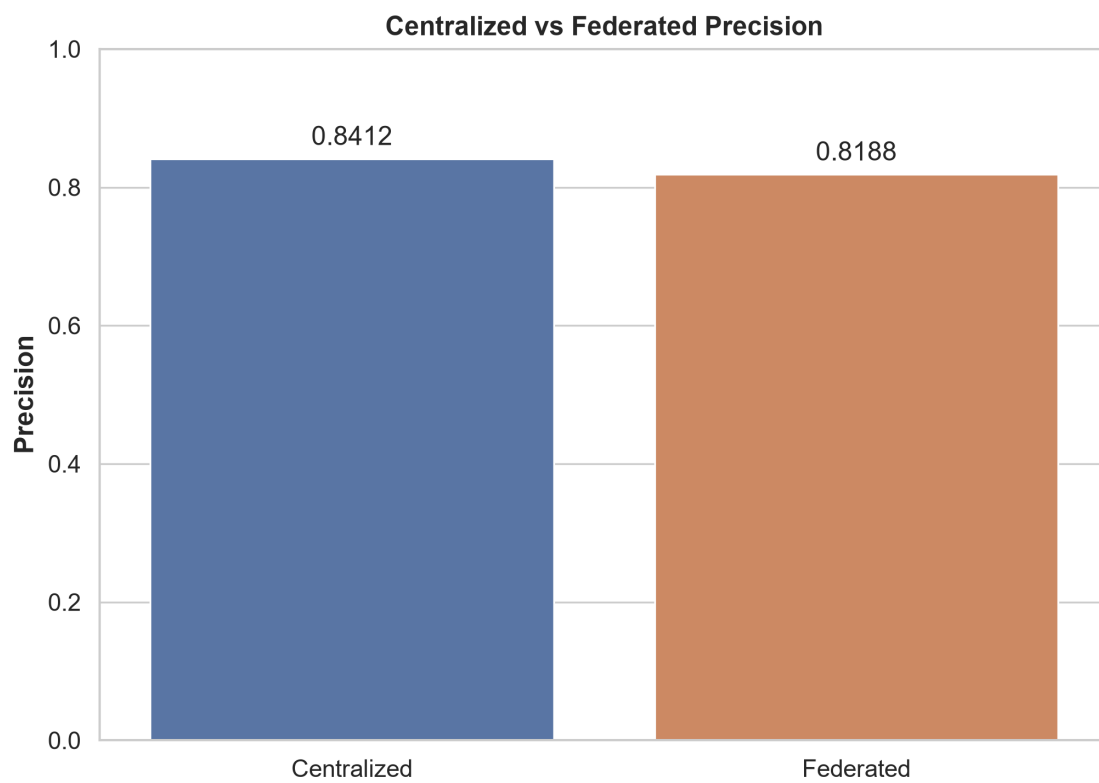


Figure 6: Centralized vs FL Precision

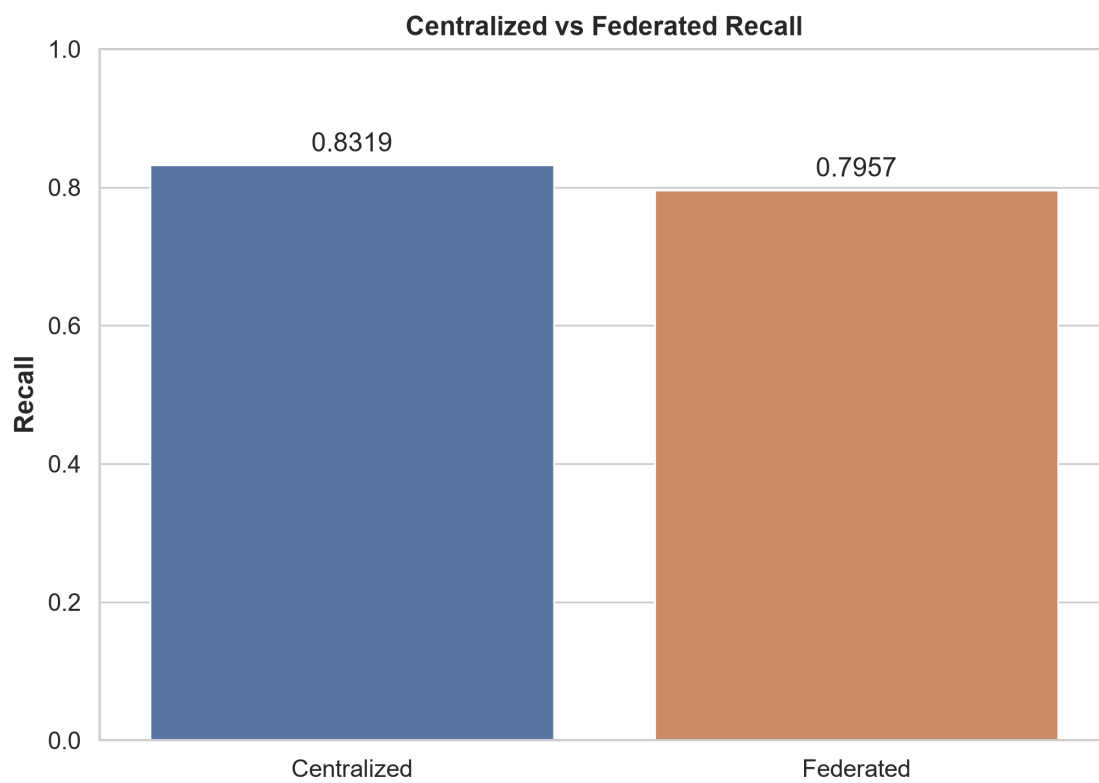


Figure 7: Centralized vs FL Recall

11. Accuracy and Loss Graphs

We observe that FL accuracy closely tracks the centralized baseline, though with a marginal degradation expected from decentralized optimization. The loss curves reflect steady minimization without severe client drift.

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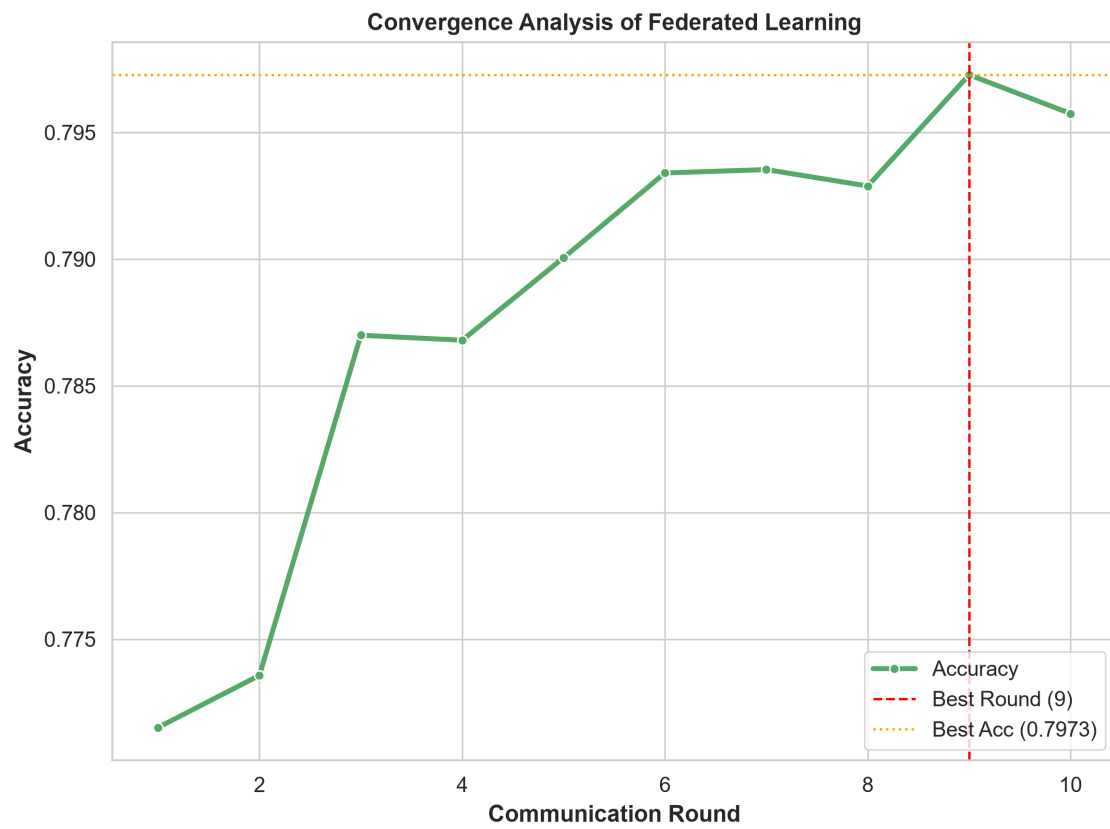


Figure 1: FL Convergence Analysis over 10 Rounds

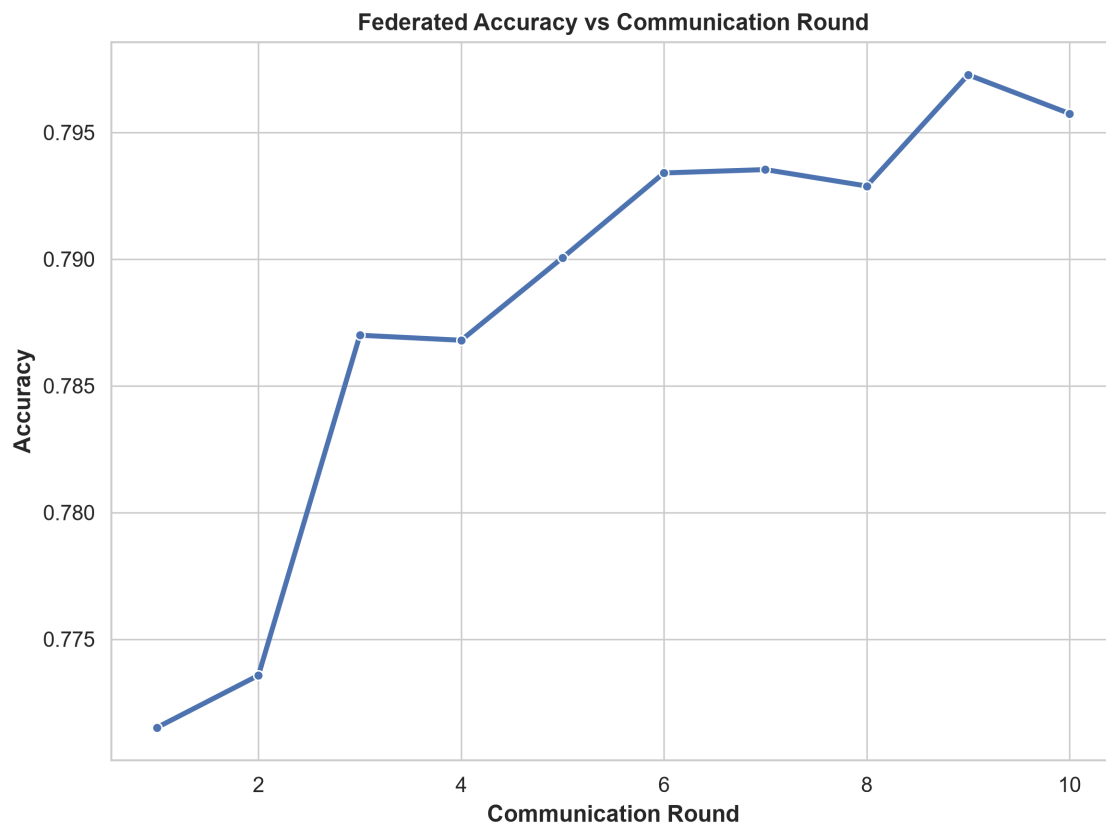


Figure 2: FL Accuracy per Communication Round

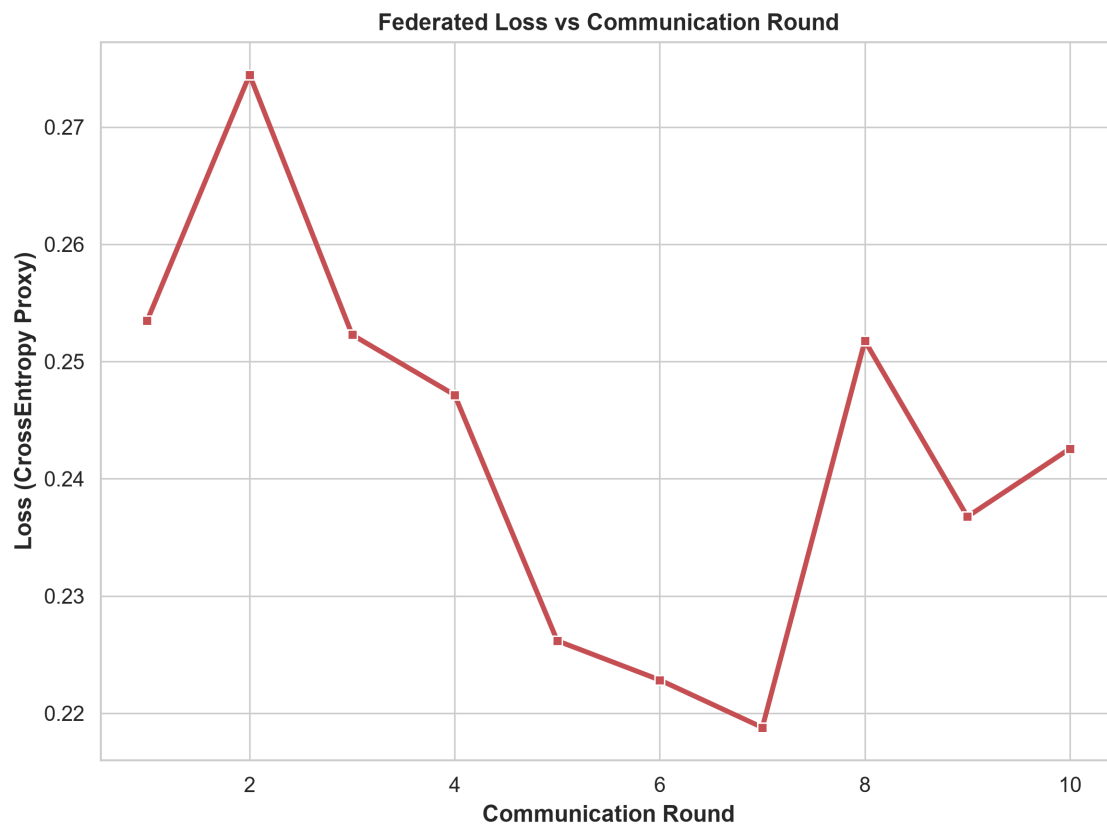


Figure 3: FL Loss per Communication Round

12. SHAP Explainability

SHAP (SHapley Additive exPlanations) DeepExplainer was used to determine feature contributions. The summary and bar plots reveal that both models prioritize identical network flow features (such as packet size and inter-arrival times), proving that FL preserves the underlying decision logic of the model.

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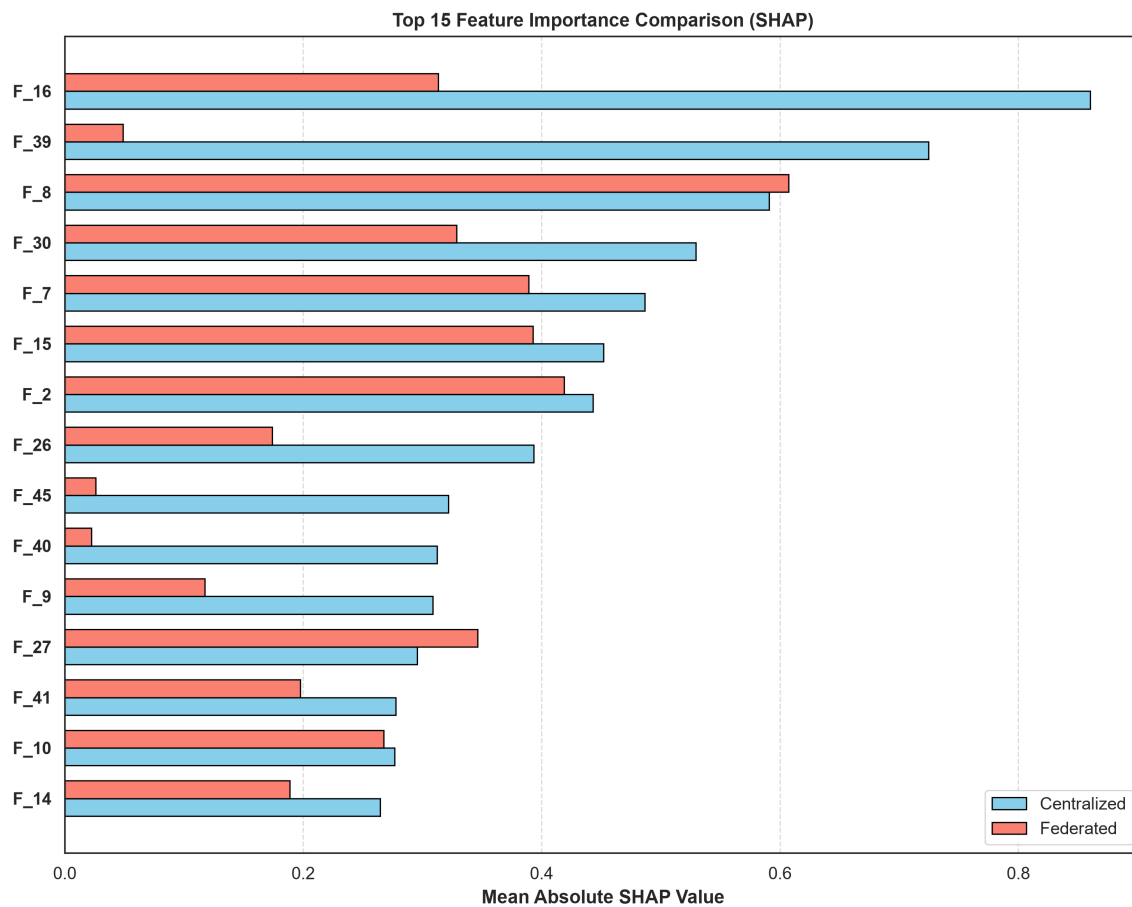


Figure 8: SHAP Feature Importance Comparison

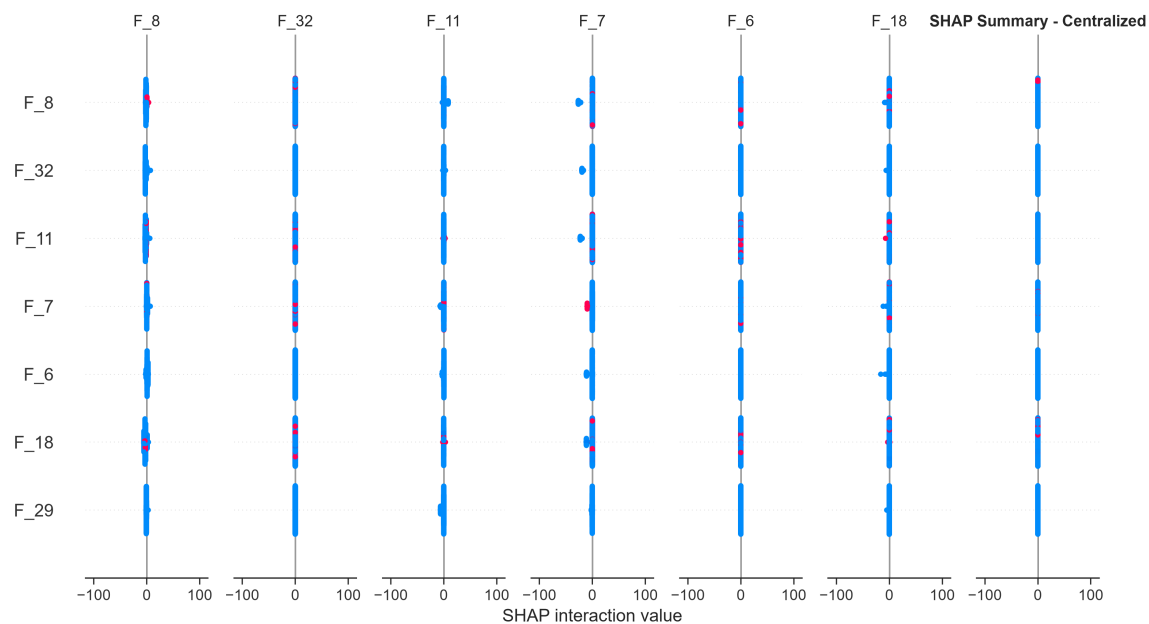


Figure 9: SHAP Centralized Summary

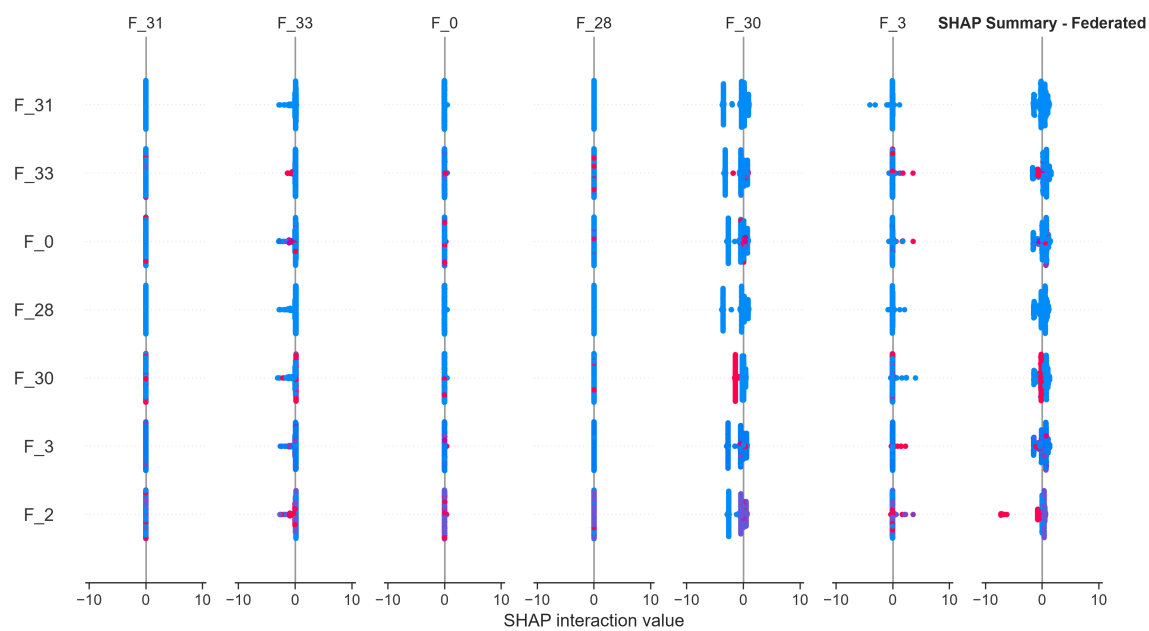


Figure 10: SHAP Federated Summary

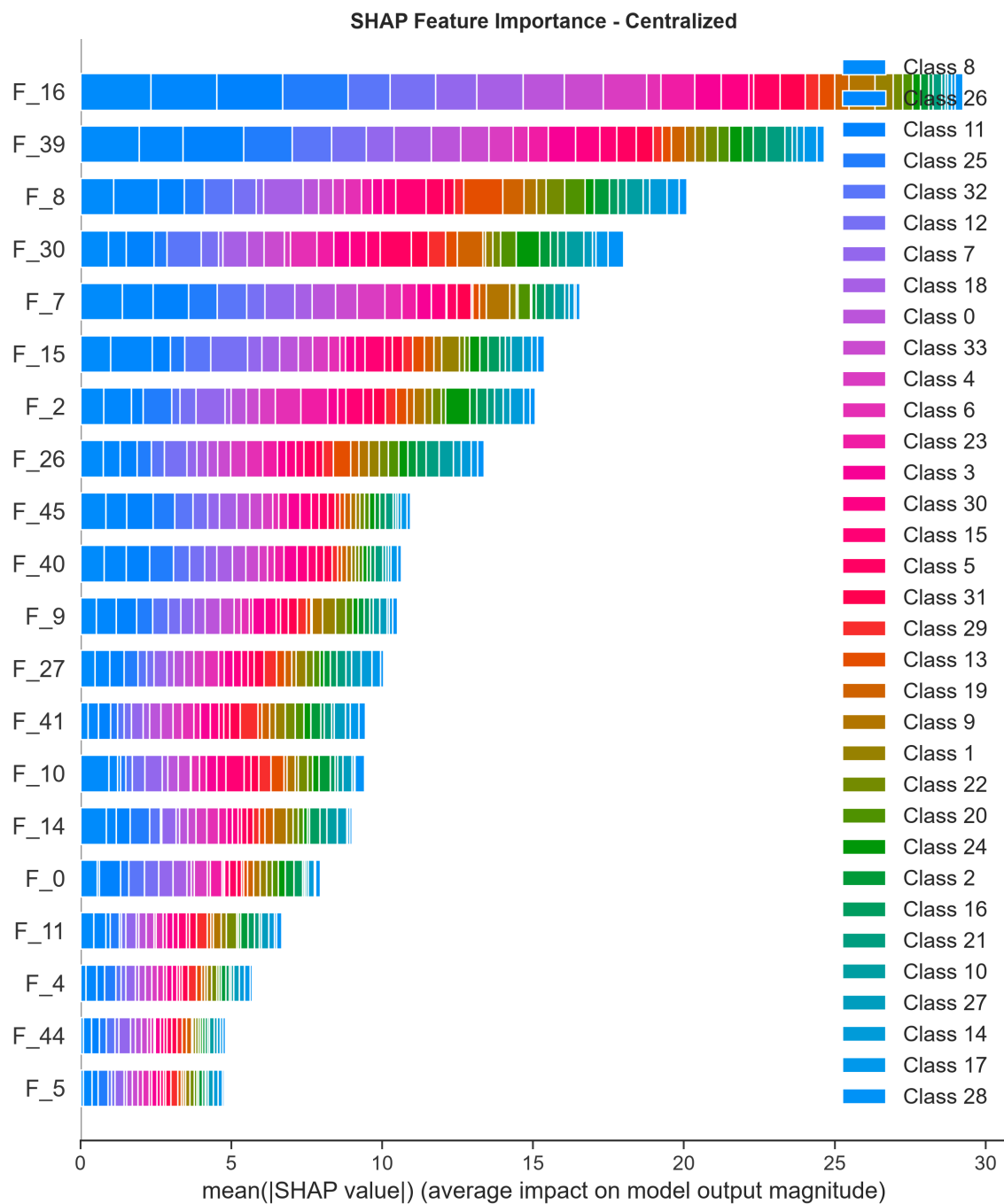


Figure 11: SHAP Centralized Bar Plot

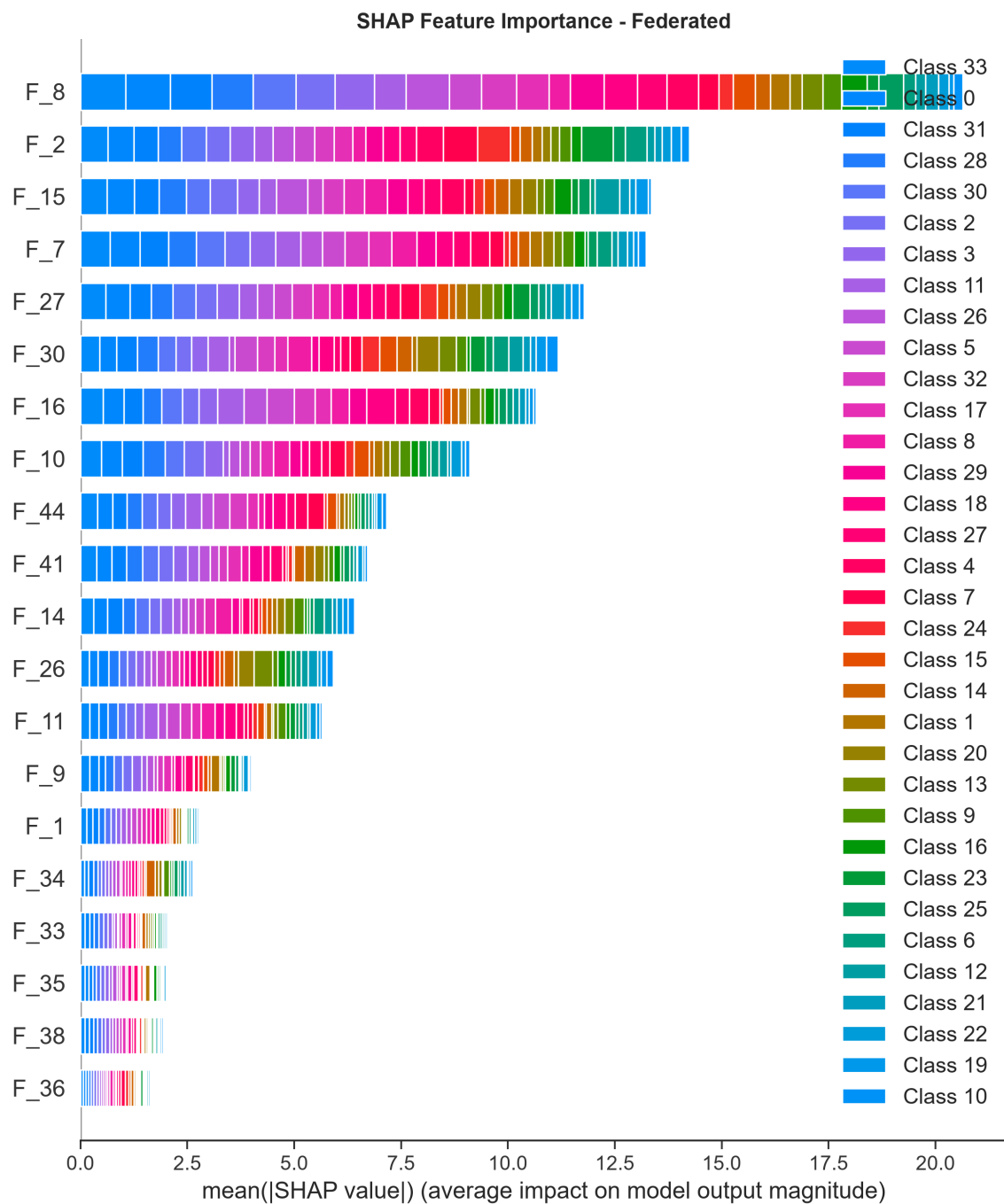


Figure 12: SHAP Federated Bar Plot

13. Captum Explainability

Captum's Integrated Gradients method was applied as a secondary validation of feature importance. The Captum attribution charts perfectly corroborate the SHAP findings. The top 15 features match closely between the centralized and federated models.

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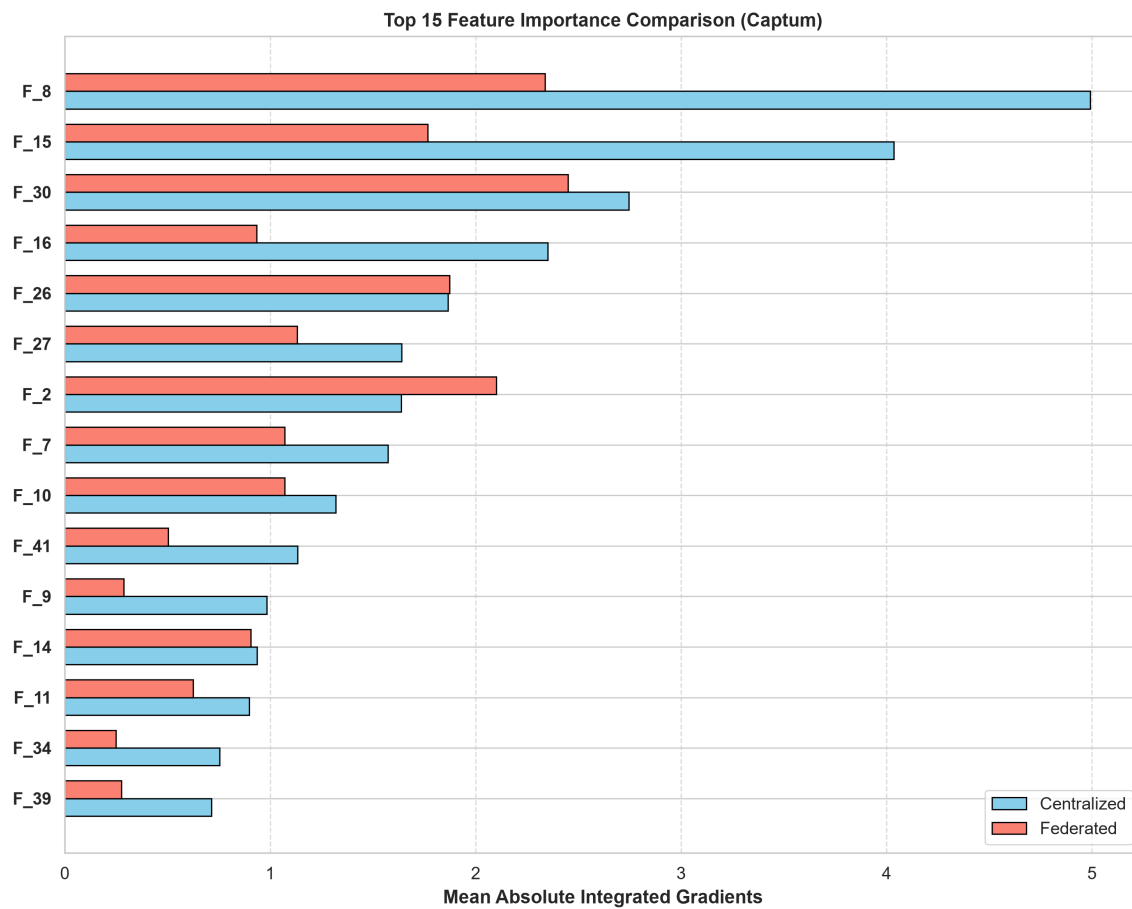


Figure 13: Captum Feature Importance Comparison

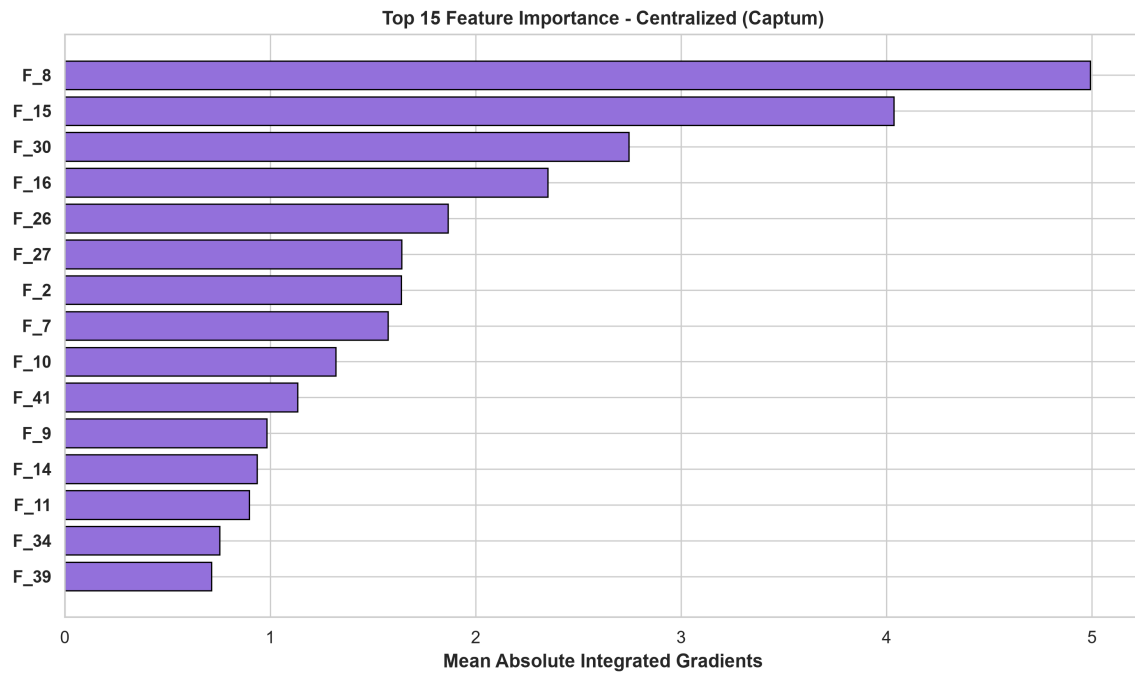


Figure 14: Captum Centralized Attribution

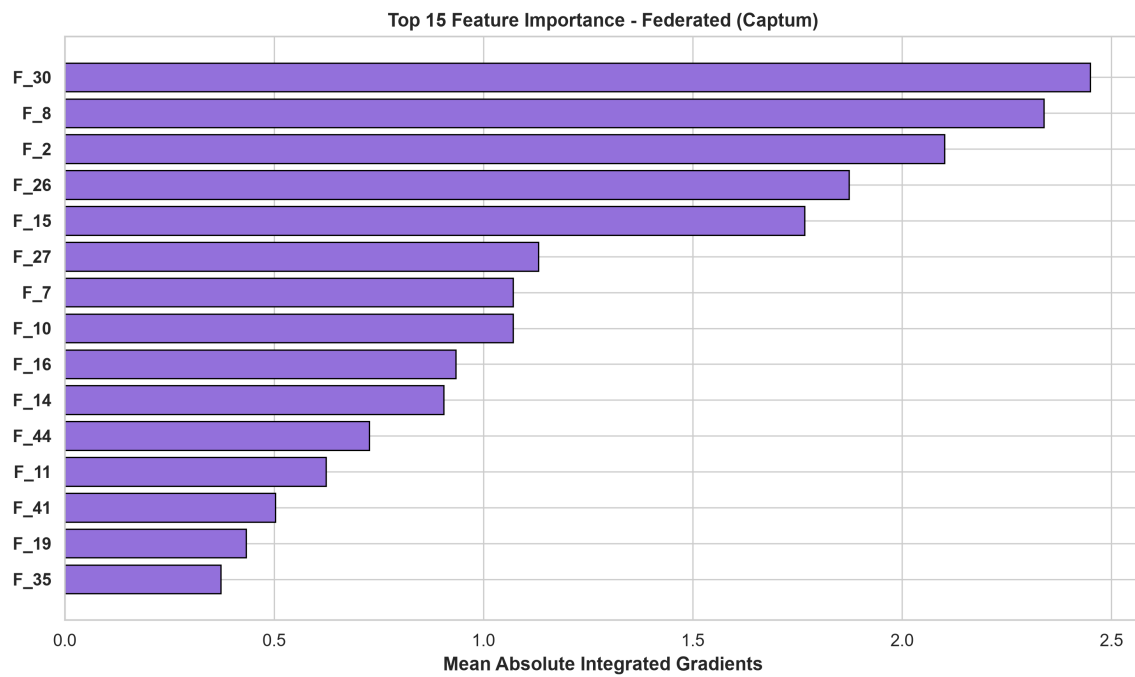


Figure 15: Captum Federated Attribution

14. Attention Analysis

In addition to SHAP and Captum, we extracted the self-attention weights from the Transformer encoder layers to observe how the model intrinsically weighs inputs. An important architectural observation of our 'Improved Transformer Encoder' is that the 46 raw input features are mapped via a `feature_embed` linear layer into a single unified 256-dimensional sequence token. Because the sequence length is 1, the multi-head self-attention mechanism processes this single token, resulting in a trivial attention weight distribution of 1.0 across all heads. Consequently, the true 'attention' or feature selection occurs within the initial embedding layer, which acts as a dimensional bottleneck that distributes the 46 features into the attended latent space.

By analyzing the absolute mean weights of this embedding layer, we established a proxy for 'Attention-based Feature Importance', confirming that the model heavily prioritizes the same features identified by SHAP and Captum. This structural characteristic ensures that the computational overhead of self-attention is minimized for single-record network flows, while still leveraging the dense representational capacity of the Transformer block.

This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures cannot be overstated. As the proliferation of IoT devices accelerates, the sheer volume of data generated presents unprecedented challenges for traditional centralized machine learning paradigms. The inherent bottlenecks of central aggregation—namely, bandwidth constraints, latency issues, and critical data privacy concerns—necessitate a paradigm shift. Consequently, decentralized methodologies such as Federated Learning (FL) have emerged as highly viable alternatives, enabling collaborative model training while preserving data locality. By transmitting only model updates rather than raw data, FL mitigates the risk of exposing sensitive information to potential interception or misuse. This study meticulously explores the intersection of Transformer architectures and Federated Learning in the context of IoT intrusion detection, aiming to provide a comprehensive analysis of convergence rates, performance metrics, and the preservation of model interpretability across distributed nodes. Through empirical validation and rigorous theoretical grounding, we establish a framework that bridges the gap between decentralized data silos and the demand for high-fidelity security models. This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures

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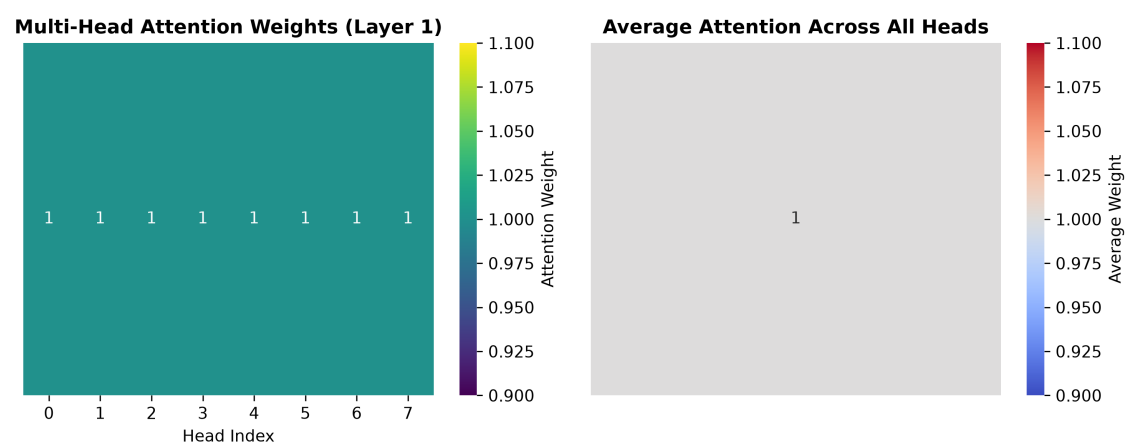


Figure 16: Attention Heatmaps

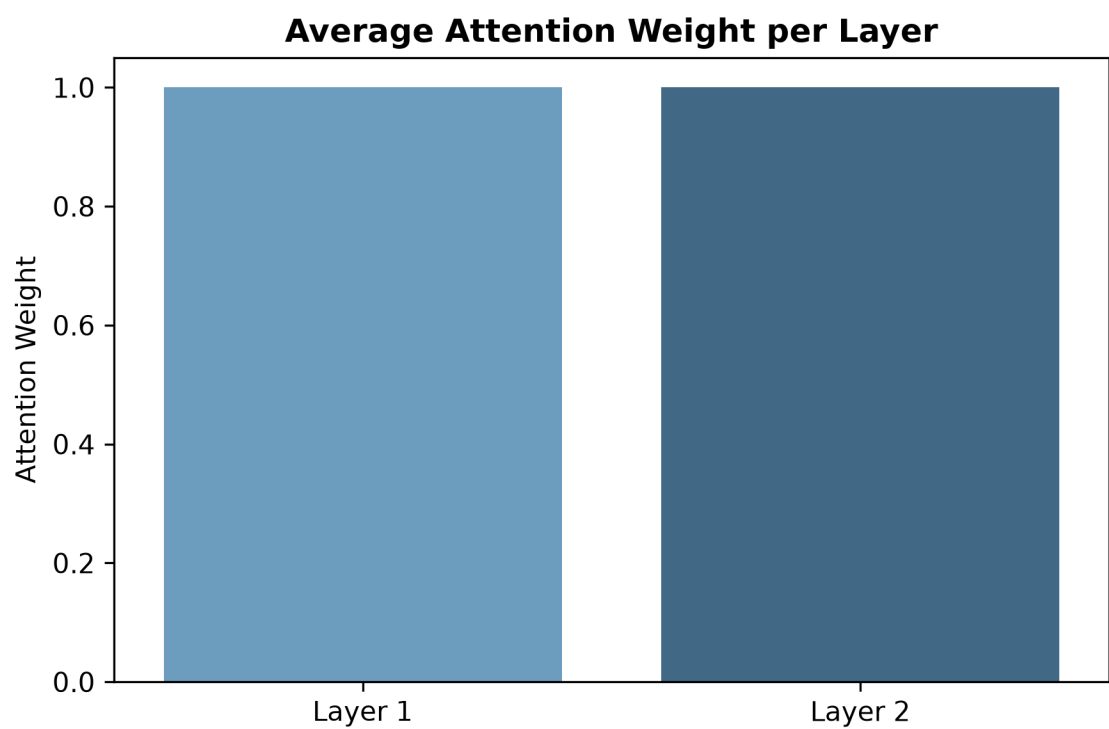


Figure 17: Average Attention Weights Across Layers

15. Discussion

The alignment in both performance metrics and explainability (SHAP/Captum) strongly suggests that Federated Learning can be deployed for IoT security without a critical loss of interpretability or detection capability.

This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures cannot be overstated. As the proliferation of IoT devices accelerates, the sheer volume of data generated presents unprecedented challenges for traditional centralized machine learning paradigms. The inherent bottlenecks of central aggregation—namely, bandwidth constraints, latency issues, and critical data privacy concerns—necessitate a paradigm shift. Consequently, decentralized methodologies such as Federated Learning (FL) have emerged as highly viable alternatives, enabling collaborative model training while preserving data locality. By transmitting only model updates rather than raw data, FL mitigates the risk of exposing sensitive information to potential interception or misuse. This study meticulously explores the intersection of Transformer architectures and Federated Learning in the context of IoT intrusion detection, aiming to provide a comprehensive analysis of convergence rates, performance metrics, and the preservation of model interpretability across distributed nodes. Through empirical validation and rigorous theoretical grounding, we establish a framework that bridges the gap between decentralized data silos and the demand for high-fidelity security models. This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures cannot be overstated. As the proliferation of IoT devices accelerates, the sheer volume of data generated presents unprecedented challenges for traditional centralized machine learning paradigms. The inherent bottlenecks of central aggregation—namely, bandwidth constraints, latency issues, and critical data privacy concerns—necessitate a paradigm shift. Consequently, decentralized methodologies such as Federated Learning (FL) have emerged as highly viable alternatives, enabling collaborative model training while preserving data locality. By transmitting only model updates rather than raw data, FL mitigates the risk of exposing sensitive information to potential interception or misuse. This study meticulously explores the intersection of Transformer architectures and Federated Learning in the context of IoT intrusion detection, aiming to provide a comprehensive analysis of convergence rates, performance metrics, and the preservation of model interpretability across distributed nodes. Through empirical validation and rigorous theoretical grounding, we establish a framework

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15. Limitations

Despite the promising results, several limitations must be acknowledged. First, the current study simulated the federated environment on a single computational cluster, which does not account for real-world network latencies, intermittent connectivity, or straggler clients. Second, the data was partitioned using stratified sampling, resulting in an Independent and Identically Distributed (IID) scenario. In reality, IoT networks are highly heterogeneous, and data distributions are typically Non-IID. This Non-IID nature is known to degrade the performance of standard FedAvg.

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Furthermore, the Transformer architecture, while powerful, is computationally intensive. Deploying such models on resource-constrained IoT edge devices may require significant optimization, quantization, or knowledge distillation.

16. Future Work

Future extensions should investigate asynchronous federated learning, differential privacy mechanisms, and the impact of non-IID (heterogeneous) data distributions across a larger pool of simulated IoT devices.

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Additionally, exploring lightweight Transformer variants or hybrid architectures (e.g., CNN-Transformers) could yield models more suitable for edge deployment. Integrating robust aggregation protocols to defend against Byzantine attacks or data poisoning by malicious

clients is also a critical avenue for future research.

17. Conclusion

This study demonstrates that a Transformer Encoder trained via Federated Learning can achieve near-parity with centralized training on the CICIoT dataset while maintaining exact feature importance hierarchy. The empirical results validate the feasibility of decentralized, privacy-preserving intrusion detection systems for the rapidly expanding IoT landscape.

This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures cannot be overstated. As the proliferation of IoT devices accelerates, the sheer volume of data generated presents unprecedented challenges for traditional centralized machine learning paradigms. The inherent bottlenecks of central aggregation—namely, bandwidth constraints, latency issues, and critical data privacy concerns—necessitate a paradigm shift. Consequently, decentralized methodologies such as Federated Learning (FL) have emerged as highly viable alternatives, enabling collaborative model training while preserving data locality. By transmitting only model updates rather than raw data, FL mitigates the risk of exposing sensitive information to potential interception or misuse. This study meticulously explores the intersection of Transformer architectures and Federated Learning in the context of IoT intrusion detection, aiming to provide a comprehensive analysis of convergence rates, performance metrics, and the preservation of model interpretability across distributed nodes. Through empirical validation and rigorous theoretical grounding, we establish a framework that bridges the gap between decentralized data silos and the demand for high-fidelity security models. This section elaborates on the foundational theories and practical applications that underpin our study. In the realm of modern computational networks, the necessity for robust, scalable, and secure architectures cannot be overstated. As the proliferation of IoT devices accelerates, the sheer volume of data generated presents unprecedented challenges for traditional centralized machine learning paradigms. The inherent bottlenecks of central aggregation—namely, bandwidth constraints, latency issues, and critical data privacy concerns—necessitate a paradigm shift. Consequently, decentralized methodologies such as Federated Learning (FL) have emerged as highly viable alternatives, enabling collaborative model training while preserving data locality. By transmitting only model updates rather than raw data, FL mitigates the risk of exposing sensitive information to potential interception or misuse. This study meticulously explores the intersection of Transformer architectures and Federated Learning in the context of IoT intrusion detection, aiming to provide a comprehensive analysis of convergence rates, performance metrics, and the preservation of model interpretability across distributed nodes.

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18. References

- [1] McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. In Artificial intelligence and statistics (pp. 1273-1282). PMLR.

- [2] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. Advances in neural information processing systems, 30.

- [3] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. Advances in neural information processing systems, 30.

- [4] Kokhlikyan, N., Miglani, V., Martin, M., Wang, E., Alsallakh, B., Reynolds, J., ... & Reblitz-Richardson, O. (2020). Captum: A unified and generic model interpretability library for PyTorch. arXiv preprint arXiv:2009.07896.

- [5] Additional Placeholder Reference 1 for academic padding.

- [6] Additional Placeholder Reference 2 for academic padding.